

What LLM Forecasters Reveal When You Ask Them About Their Uncertainty: A Per-Family, Cross-Corpus Study

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Abstract

LLM forecast calibration is usually reported at the panel-aggregate stratum, which hides where the per-model signal actually lives. We elicit five uncertainty primitives in the same prompt (worry scalar, bid-ask spread, self-predicted Brier interval, reasoning trajectory, reference-class outside view) from five families (Claude Opus 4.7, GPT-5.5, GPT-5.4-mini, Gemini 2.5 Flash, DeepSeek Chat) on a public-domain $N=42$ corpus, with every claim resolved through three legal verdicts (**h1_supported**, **h0_kept**, **inconclusive_underpowered**) and required sample sizes pre-computed via Fisher- z power. Four claims survive in scoped form: (1) the per-model worry-Brier relationship is direction-flipped between families and replicates across three independent pilots ($N=264$); (2) **cognitive content and calibration are decoupled — a quantified Chain-of-Thought unfaithfulness finding**: within-family Spearman correlation between rationale failure-mode word count and per-row Brier is near zero for two of five families (Gemini $\rho=-0.045$, DeepSeek $\rho=+0.049$) at $N=60$ where $|\rho|\geq 0.36$ is required for power, the model produces failure-mode reasoning text on demand but does not recompute the point estimate from that content; (3) a multi-channel OLS on the initial public corpus generalizes for one of five families under leave-one-out cross-validation (GPT-5.5, $R^2_{\text{LOO}}=+0.31$; the other four are at or below zero), but a later $N=157$ per-family v28a re-audit finds no positive channel-only LOO R^2 after shortcut controls, so this is diagnostic rather than deployable; (4) a reference-class outside view, the model first lists five similar resolved questions and emits their historical YES rate before its own probability, partially replicates cross-corpus, with $\rho=+0.39$ **h1_supported** for GPT-5.4-mini, beating that family’s own point estimate ($\rho=+0.21$). The bid-ask spread and self-predicted-Brier interval are introduced as novel elicitation primitives; probabilistic forecasting is the domain where Chain-of-Thought unfaithfulness can be mathematically quantified against ground-truth Brier rather than human judgement [7]. The audit pipeline retracted four prior-program “null” findings to **inconclusive_underpowered** and five cross-corpus claims to corpus-specific; we report the retractions transparently. The operational lesson is diagnostic rather than deployment-grade: per-family elicitation surfaces can reveal error, and later source-balanced contrastive packets support bounded pairwise ranking plus an experimental graph-derived probability translation with one-direction cross-packet transfer, but a tiny joined-market slice is only directionally favorable for the translated layer and production policy remains unvalidated.

1 Introduction

Eigenproblem. Which LLM forecasting behaviours are structural inheritances of the training-text distribution the model samples from, and along which axes can the transfer pattern be predicted ex ante from the structural origin of each behaviour?

Forecasting measurement order. For forecasting, the inheritance question is not the first operational question. A black-box LLM forecaster first has to know whether the row is valid forecast evidence for its generation: was the resolved answer already source-visible, and is the benchmark measuring prediction rather than retrieval or source familiarity? Conditional on that, the next question is which emitted channel is reliable for this family/source cell: point estimate, worry scalar, bid-ask spread, self-predicted Brier interval, reference-class base rate, or cross-family disagreement. Only after those two questions does intervention policy become meaningful: does any prompt, carrier, abstention, review, or routing rule improve Brier or utility against explicit controls? The applied program is therefore organized by evidence function: validity gates decide what can count; calibration transforms probabilities; ranking tests relative judgment; allocation decides when to buy outside information; external baselines decide whether LLMs add value; inheritance theory explains transfer only after the measurement surface is valid. The three-axis inheritance frame below explains why some behaviours and bias surfaces transfer; the forecasting spine of the paper is source-validity, family/source channel reliability, and controlled policy translation.

Epistemic shape. The paper is empirical first and theoretical second. We ran a multi-pilot forecast-calibration program (five-model panel, two corpora, 30+ measured findings, pre-registration discipline) without a unifying theoretical commitment. The three-axis inheritance frame below was **induced from the body of measurements in retrospect** as the most parsimonious structure organising the per-family heterogeneity, the universal patterns, and the bias-transfer findings together. We treat the framework’s status accordingly: it earns the right to be called a unifying frame only to the extent that it makes pre-registered predictions on bias transfers we had not yet measured at the time of synthesis. Section §15 reports the in-sample evidence; section §17 reports the predictive test on a held-out slate of biases pre-registered under the framework.

The three-axis inheritance frame (inductively derived). The findings in this paper are organised by three structural axes along which training-text inheritance shows up in the forecast distribution. Each axis governs a different family of measurements; together they cover the program’s per-family heterogeneity, universal patterns, and bias-transfer findings.

- **Axis 1, Elicitation surface.** Which channels the model exposes when asked to forecast (a point estimate p_{success} , a bid-ask spread $p_{\text{buy yes max}}/p_{\text{sell yes min}}$, a tail-worry scalar, a self-predicted Brier interval, an asymmetric-loss decision threshold). The model emits the channels that its training distribution rewards; channels differ across families because training mixes differ.
- **Axis 2, Bias-mechanism class.** A three-cell taxonomy on which human cognitive biases transfer to LLM forecasting and how. *Pure heuristic* biases (cognitive shortcuts with high natural-speech footprint, like anchoring or self-overconfidence) INHERIT. *Pure motivational* biases (intrinsic utility / fear / consumption preference, like loss aversion or hyperbolic discounting) ESCAPE: the model has no utility function, the text has no representation of the mechanism, no transfer. *Systemic motivational* biases (intrinsic utility but heavy case-study representation in training text, like sunk cost or in-group framing) MIMIC: the model emits utility-like behaviour because the text frequency of the bias being discussed drives the output, not a utility function operating.
- **Axis 3, Family/alignment overlay.** Family identity reshapes inherited-bias magnitude and per-family channel surfaces. Per-family heterogeneity in worry-channel sign, channel decomposition R^2 , and meta-cognitive routing all live on this axis. We call this an alignment overlay only as a hypothesis for matched-checkpoint follow-up; the measurements here are cross-family and therefore confound RLHF/alignment with pretraining mix.

The contribution shape is the **synthesis** of the three axes plus the MIMIC category as a specifically novel articulation on axis 2: prior LLM-bias work has not separated transfer-because-mechanism-is-shared (inherit) from transfer-because-bias-is-discussed-extensively-in-text (mimic). The MIMIC category lets the same surface observation carry different interpretations depending on which cell the bias belongs to. A later 180-call anti-bias-collapse smoke scoped down the strongest version of the prompt-actuation mechanism: directional MIMIC collapse did not survive label-shuffle control, and raw-gap adjustment reversed the MIMIC coefficient. We therefore keep MIMIC as a representational taxonomy, not as a clean anti-bias-prompt law.

Methodological backbone. A power-aware verdict resolver (Fisher- z sample-size computation before fire, three legal verdicts after `h1_supported` / `h0_kept` / `inconclusive_underpowered`, equivalence testing instead of “ $p > 0.05$ ” as a no-effect claim, BH-FDR across panel tests, leave-one-out R^2 at small N) sits underneath every empirical claim in the paper. Internally, 8 of 12 prior nulls in this program turned into `inconclusive_underpowered` under the resolver, and 5 cross-corpus claims retracted to corpus-specific. The discipline ITSELF is a methodological contribution because the same overcalling failure mode plausibly affects the field’s existing $N=5-20$ cross-corpus headlines. A second-order application (corpus-validity drift: when a benchmark’s resolution dates are post-cutoff for the publication-era LLMs and pre-cutoff for the next-generation LLMs replicating against it) is the most consequential field-corrective the discipline has produced, applied to the Halawi 2024 dataset [3] in §18.

Reading order. Axes 1–3 are introduced in §3; Axis 1 findings sit in §7, §8, §11; Axis 2 findings sit in §15 (with the MIMIC category as the central original contribution) and inside the universal-patterns diagnostic rules in §14 (the confident-NO discount as the INHERIT cell in policy-hypothesis shape); Axis 3 findings sit in §5, §10, and the per-family channel-routing results. The methodological backbone runs underneath all of them and is collected in §18. An honest scope statement (§19) and the reproducibility plan (§20) close the paper.

2 Setup

2.1 Five LLM families

Claude Opus 4.7 (Anthropic, via Claude Code CLI), GPT-5.5 and GPT-5.4-mini (OpenAI, via Codex CLI), Gemini 2.5 Flash (Google GenAI API), DeepSeek Chat (DeepSeek API). All API calls temperature=0 unless explicitly noted. All paths audited for web-search-tool access; all prompts include an explicit “do NOT use web search” instruction.

2.2 Two corpora, four confounded axes

We use a *low-overlap corpus* of $N=15$ bespoke questions drawn from a research-internal domain with effectively zero LLM-pretraining overlap (formal-proof tactic combinations, gated state transitions in a substrate-specific apparatus, ground-truth-confirmed forecasting contracts), and a *public-domain corpus* of $N=42$ contracts from prediction markets (Manifold, Polymarket), stock-close thresholds, and ETF moves.

The four axes are confounded in the current data. Low-overlap-corpus questions are long-complex-alien-no-public-prior *together*; public-domain are short-simple-familiar-with-public-prior *together*. Our cross-corpus result (§9) is therefore evidence that *at least one* of these axes matters; a

Axis	Low-overlap ($N=15$)	Public-domain ($N=42$)
LLM pre-training overlap	Effectively zero	Substantial
Question length	247 chars (35 words)	99 chars (17 words)
Question structure	Multi-conditional, code/math symbols, parenthetical embedding	Atomic, simple syntactic structure
Implicit base rate	Ill-defined (bespoke operations)	Public-market priors well-trained

4-cell de-confounded corpus design that breaks the confound is the highest-priority methodological follow-up (§19).

The low-overlap corpus is research-internal and not publicly releasable in raw form; the reproducibility plan (§20) addresses this through a sanitized release with neutral identifiers and a parallel public-but-niche-academic corpus matching the four-axis profile.

2.3 Pre-registration and power-aware verdicts

Every pilot is pre-registered in a SQLite `pre_registrations` table before launch with hypothesis, null, target effect size, n_{required} (auto-computed via Fisher- z at $\alpha=0.05$ two-tailed, 80% power, Spearman correction +6%), falsifiers, and pass-gate. Resolution uses three legal verdicts:

- **h1_supported**: 95% CI on observed ρ excludes 0.
- **h0_kept**: 95% CI wholly within $(-\text{target}, +\text{target})$; data rules out a meaningful effect.
- **inconclusive_underpowered**: observed $|\rho|$ below detectability at the run N ; CI wide.

Brier-delta tests use paired-contract sign-flip permutation with a 90% bootstrap CI plus a BIC-approximation Bayes factor. “No effect” claims also require TOST equivalence at a pre-stated bound. Power thresholds at $\alpha=0.05$ / power= 0.80: $N=15 \Rightarrow |\rho| \geq 0.68$; $N=42 \Rightarrow 0.43$; $N=91 \Rightarrow 0.30$.

3 The three-axis inheritance frame

Before reporting the empirical findings, we name the structural axes the program’s measurements turned out to organise along. These axes were not stated in advance; they are the most parsimonious reading of the body of findings the program produced. The framework’s predictive content is tested separately (§17).

Axis 1: elicitation surface. An LLM forecaster, when asked, emits a set of channels: at minimum a point estimate p_{success} ; optionally a tail-worry scalar (“how worried are you?”), a bid-ask spread ($p_{\text{buy yes max}}, p_{\text{sell yes min}}$), a self-predicted Brier interval ($b_{\text{lo}}, b_{\text{hi}}$), an asymmetric-loss decision threshold under specified false-positive vs. false-negative cost regimes, or an outside-view base rate from analogous prior contracts. The set of channels the model surfaces is itself the first inheritance signature: training-mix differences across model families produce different elicitation surfaces, even when prompted identically. §7 introduces the bid-ask spread as a novel diagnostic error-warning channel that is elicitable on the low-overlap corpus; §8 shows the three uncertainty channels are statistically independent on a paired panel, motivating per-family channel analysis; §11 shows only one family generalises a multi-channel decomposition under leave-one-out cross-validation. All three findings sit on this axis.

Behavioral latent-carrier interpretation. We do not observe the hidden activations of the closed models used here, so our channel claims are behavioral rather than mechanistic-interpretability claims. Still, the right unit is closer to a structured intermediate representation than to the generated rationale. Recent latent-prediction theory shows that learning or predicting internal representations can avoid the sample-complexity cost of token-level prediction on hierarchical data [6]. Our setting is not training-time latent prediction, but the analogy is operationally useful: uncertainty channels and source-bound evidence fields can be scored against outcomes, while free-form rationale text is a noisy surface artifact. In two small follow-up smokes across two families, structured evidence carriers beat free prose on mean Brier, while the stricter two-step variant did not consistently beat a same-turn carrier. The combined means were baseline 0.171038, free prose 0.146103, two-step carrier 0.103278, and same-turn carrier 0.098425. A later placebo-control smoke was harsher: among the 30 schema-valid rows, baseline mean Brier was 0.078000, two-call prose 0.107254, same-turn carrier 0.110300, free prose 0.122767, and two-step carrier 0.149921; ten additional Codex rows failed at runtime before emitting forecasts. This is not a validated law. It keeps structured carrier fields as a hypothesis for larger paired tests, but it argues against treating the stronger two-step break as an established mechanism.

Axis 2: bias-mechanism class. We propose a three-cell taxonomy for how human cognitive biases transfer to LLM forecasting.

- **PURE HEURISTIC** (cognitive shortcut, high natural-speech footprint). Examples: anchoring [14], self-overconfidence [8], the gambler’s fallacy [13]. Predicted manifestation: **inherit** (training-text frequency drives the LLM output directly via the token distribution).
- **PURE MOTIVATIONAL** (intrinsic utility / fear / consumption preference; low representation in raw analytical text). Examples: loss aversion in the Kahneman-Tversky sense [5], cumulative prospect theory probability weighting [15], hyperbolic discounting [2]. Predicted manifestation: **escape** (the LLM has no utility function and the text has no representation of the bias mechanism, so no transfer).
- **SYSTEMIC MOTIVATIONAL** (intrinsic utility BUT heavy case-study / behavioural-economics-literature representation in training text). Examples: sunk-cost reasoning [1], endowment / status-quo bias [12], in-group framing [10]. Predicted manifestation: **mimic** (the LLM emits utility-like behaviour, but the mechanism producing the output is text-frequency representation of the bias being discussed in training text, not a utility function operating).

The MIMIC cell is the central original contribution of the framework. Prior LLM-bias work has not separated transfer-because-the-mechanism-is-shared (INHERIT) from transfer-because-the-bias-is-discussed-extensively-in-text (MIMIC). The cell distinction matters because it suggests different intervention surfaces, but the direct anti-bias-prompt signature did not promote as a clean law: in the 180-call collapse smoke, MIMIC mean collapse was positive but underpowered, the class-label shuffle was null ($p = 0.5387$), and the raw-gap-adjusted MIMIC coefficient was negative (-0.076587 , $p = 0.0025$). A no-call matched audit also found poor existing support overlap: at raw-gap caliper 0.05, within-family matching left only 16 with-replacement / 15 no-replacement pairs and flipped the MIMIC-minus-control collapse estimate negative (-0.072750 , $p = 0.0008$; greedy no-replacement -0.077561 , $p = 0.0006$), while the no-caliper positive estimate relied on large raw-gap distances. The supported claim is therefore taxonomic and diagnostic; a causal anti-bias-collapse mechanism requires new matched raw-gap strata or randomization.

Axis 3: alignment overlay. RLHF and post-training alignment damp some inherited biases (truthfulness, helpfulness training reduces obvious factual hallucinations and overt in-group preference) and reshape per-family channel surfaces. Per-family worry-Brier sign differences (§5), per-family channel-decomposition R^2 differences (§11), and the meta-cognitive routing failure where three of five families pick the WRONG channel they should rely on at $p \leq 0.005$ each (§12) all live on this axis. The empirical signature is that family identity, conditional on channel and contract, is itself a central predictor.

The methodological backbone is orthogonal to the three axes. A verdict-resolver discipline (Fisher- z power calculation before fire, three legal verdicts after, equivalence testing, BH-FDR, LOO- R^2 at small N) sits underneath every axis-1/2/3 measurement. The discipline produced the program’s most consequential retraction: 8 of 12 prior “no effect” claims turned into **inconclusive_underpowered** under the resolver, and the corpus-validity-drift filter (`resolve_date > max(panel_cutoff)`) empties the most-cited 2024 LLM-forecasting benchmark for the current LLM generation (§18).

4 Foundation: the worry scalar is Bayesian-coherent tail risk

The basic uncertainty primitive used throughout this paper is `tail_insurance_premium` (worry scalar, 1–100, requested in the same prompt as the binary forecast). This primitive was validated in a separate within-program pilot before any of the downstream analyses: at pool level $N=590$ on an independent cross-pool forecast corpus, the worry scalar is a Bayesian-coherent tail-risk signal, more-worried forecasts ARE more likely to be wrong, in expectation. This is the only well-powered *positive* pool-level validation of a verbalized-uncertainty primitive in our program; everything that follows uses worry as a base channel, and the per-model decomposition in §5 shows the pool-level signal is the average of opposite per-family signs.

5 Per-family worry-Brier direction split, three-pilot replication

For each LLM family we compute the rank correlation between the worry score and the realized squared error (Brier) of the forecast that the worry accompanied. Three independent pilots ($N=164$, $N=85$, and $N=115$; cumulative $N=264$) replicate this analysis:

- Claude Opus 4.7: $\rho < 0$, most worried when most right (anti-Dunning-Kruger).
- GPT-5.5: $\rho < 0$, same direction.
- GPT-5.4-mini: $\rho > 0$, direction-sensible (worried when wrong).

Per-pilot detectable $|\rho|$ at 80% power is ≈ 0.30 for $N=85$; observed $|\rho|$ values span 0.31–0.56 across families, clearing detectability for Claude and GPT-5.5 individually. Cross-pilot Stouffer Z meta-analysis combines the three pilots at combined $p < 0.01$. **This is the per-model finding that survives the largest paired-permutation budget in the program.**

A natural reviewer hypothesis is that the sign-flip is mediated by task complexity: complex tasks produce longer rationales; longer rationales correlate with both higher stated worry AND lower Brier (more thorough reasoning); the apparent inversion is an artifact of this joint mediation. We test the mediator directly on $N=45$ –91 per family. None of the five families exhibits the predicted pattern of “long rationale $\rightarrow$ high worry AND long rationale $\rightarrow$ low Brier.” Claude’s $\rho(\text{rat_length}, \text{tip}) \approx 0$

(claude does not worry more about long rationales); Gemini’s $\rho(\text{rat_length}, \text{Brier})$ is POSITIVE, the opposite of the reviewer’s predicted direction. **The task-complexity-via-rationale-length mediator is rejected.**

The refined mechanism we propose is given in §10 after §6 provides the upstream decoupling result.

6 Cognitive content and calibration shift are decoupled

Across three pilots (pooled $N=320$ analytic), we measure the within-family Spearman correlation between the count of failure-mode words in the rationale and the per-row Brier:

Family	N	$\rho(\text{failure-mentions}, \text{Brier})$
Claude Opus 4.7	≈ 60	+0.27
GPT-5.5	≈ 60	+0.18
GPT-5.4-mini	≈ 60	+0.21
Gemini 2.5 Flash	60	−0.045
DeepSeek Chat	59	+0.049

At $N=60$ per family the detectable $|\rho|$ at 80% power is ≈ 0.36 . Gemini and DeepSeek’s near-zero ρ values fall within the **h0_kept** band at $|\rho| < 0.20$. The model produces failure-mode reasoning text on demand, cognitive shift is real, but does not recompute the point estimate from that content. This is the cleanest mechanism evidence in the program and connects to the chain-of-thought-faithfulness lineage [7]; the original trio shows weak positive correlation consistent with partial propagation rather than full decoupling.

7 Bid-ask spread: a novel meta-uncertainty primitive

Instead of asking the model for one probability, we elicit three in the same prompt: the highest probability at which it would BUY a YES contract, its point estimate, and the lowest probability at which it would SELL YES. The spread (sell − buy) is the model’s self-reported uncertainty about its own probability, a Knightian bracket distinct from the unidirectional worry scalar.

Family	Mean spread	$\rho(\text{spread}, \text{Brier})$	95% CI (Fisher- z)	Verdict
Claude Opus 4.7	0.21	+0.49	[−0.03, +0.80]	CI barely touches 0
GPT-5.5	0.24	−0.05	[−0.55, +0.47]	inconclusive
GPT-5.4-mini	0.26	+0.69	[+0.28, +0.89]	h1_supported, $p=0.013$
Gemini 2.5 Flash	0.21	+0.40	[−0.15, +0.75]	inconclusive
DeepSeek Chat	0.26	+0.69	[+0.27, +0.89]	h1_supported, $p=0.013$

Pre-registered pass-gate met: two of five families clear $\rho > 0.3$ with non-zero-crossing CI at $N=15$ low-overlap. No family shows the wrong sign. To our knowledge no prior LLM-forecasting paper has elicited bid-ask spread as an uncertainty primitive. The market-microstructure framing forces the model into a more demanding cognitive task than the scalar worry, which we conjecture is exactly why it produces a more honest meta-uncertainty signal (the elaboration is supported by the channel-orthogonality result in §8).

Cross-corpus failure-to-replicate. On the public-domain $N=42$ corpus, all per-family CIs cross zero (detection threshold $|\rho| \geq 0.43$ at 80% power for $N=42$). The point-estimate ρ values

for GPT-5.4-mini (-0.11) and DeepSeek (-0.07) flip sign. Either (a) the primitive is corpus-specific consistent with the four-axis confound, or (b) $N=42$ remains underpowered to confirm a smaller-but-real positive effect. To distinguish requires $N \geq 91$ public-domain. The honest framing is *low-overlap-corpus positive with explicit cross-corpus failure-to-replicate*.

8 LLM uncertainty channels are statistically independent

Pooling three low-overlap-corpus full pilots (worry-scalar baseline, bid-ask spread, reasoning-trajectory variance) on the $N=75$ rows where all three are observed, the pairwise Pearson correlation matrix among the three uncertainty channels:

	worry scalar	bid-ask spread	trajectory variance
worry scalar	1.00	+0.08	+0.17
bid-ask spread	+0.08	1.00	+0.03
trajectory variance	+0.17	+0.03	1.00

The off-diagonals are all below 0.20. PCA would not produce dimensionality reduction. Each primitive is measuring something genuinely different about how a model expresses uncertainty.

Per-family $\rho(\text{channel}, \text{Brier})$ reveals that each model family has a different “best channel”:

Family	$\rho(\text{worry}, \text{Brier})$	$\rho(\text{spread}, \text{Brier})$	$\rho(\text{traj_var}, \text{Brier})$	Best channel
Claude Opus 4.7	-0.38 (sign-flipped)	+0.02	+0.08	worry (with sign flip)
GPT-5.5	+0.07	+0.12	-0.18	weak across all
GPT-5.4-mini	-0.18	+0.55	-0.22	bid-ask spread
Gemini 2.5 Flash	+0.10	+0.10	+0.25	trajectory variance
DeepSeek Chat	-0.26 (sign-flipped)	+0.29	-0.10	bid-ask spread

There is no universal best uncertainty primitive across LLMs. The diagnostic recipe is per-family: identify which channel carries error information for each family before attempting any policy translation. Per-primitive per-family single-channel R^2 on the same-pilot Brier is 10–26% for the families where the channel works (GPT-5.4-mini’s bid-ask explains 26% of bid-ask-pilot Brier; DeepSeek’s 19%). The multi-channel R^2 from the all-channels-in-one-prompt pilot is reported with leave-one-out cross-validation in §11, where only GPT-5.5 survives LOO on the initial public-domain corpus; the larger v28a refill audit later demotes this to a diagnostic surface rather than a general classifier.

Scope of the orthogonality claim. The result is at the *expression* level. A reasoning-decomposition claim would require mechanistic interpretability tooling, logit lens, attention probing, causal intervention on chain-of-thought spans, which we do not use. Our primitives are black-box behavioral probes of uncertainty expression; any per-family routing policy remains prospective and must clear heldout Brier controls. The mechanistic-reasoning claim is left for white-box work.

9 Frequency framing transfers in a corpus-specific way

Tetlock-style “imagine 10 similar contracts; how many resolve as success?” [11] framing on the low-overlap corpus ($N=15$ each, paired permutation vs the same-corpus baseline):

On the public-domain $N=42$ corpus:

Family	Δ -Brier	95% CI	p (perm)	Verdict
Claude Opus 4.7	-0.006	$[-0.05, +0.03]$	0.85	inconclusive
GPT-5.5	+0.014	$[-0.06, +0.09]$	0.74	inconclusive
GPT-5.4-mini	-0.118	$[-0.21, -0.04]$	0.013	h1_supported
Gemini 2.5 Flash	-0.118	$[-0.28, +0.03]$	0.19	inconclusive
DeepSeek Chat	+0.115	$[-0.02, +0.28]$	0.18	inconclusive

Family	Δ -Brier	95% CI	Verdict at bound 0.05
Claude Opus 4.7	-0.003	$[-0.030, +0.028]$	h0_kept
GPT-5.5	+0.005	$[-0.012, +0.023]$	h0_kept
GPT-5.4-mini	+0.003	$[-0.042, +0.053]$	borderline
Gemini 2.5 Flash	+0.052	$[+0.000, +0.112]$	not h0_kept ($p=0.071$)
DeepSeek Chat	-0.001	$[-0.031, +0.030]$	h0_kept

GPT-5.4-mini’s low-overlap-corpus improvement does NOT replicate on the public-domain corpus. Direction goes from -0.118 to $+0.003$. For 3 of 5 families the public-corpus Brier-delta is **h0_kept** at bound ± 0.05 , the data actively rules out a 5%-Brier effect on public-domain questions. Sub-source stratification within the public corpus reveals heterogeneity the aggregated null masks: Gemini on Polymarket-style questions trends $\Delta = +0.138$ ($p=0.14$); DeepSeek shows opposite-sign effects on Manifold-bulk vs Polymarket; GPT-5.4-mini is genuinely corpus-specific (every sub-source $\Delta \in [-0.002, +0.007]$, no false-negative from aggregation).

The clean claim. Frequency framing’s effect on LLM forecasters is corpus-specific and model-specific: it improves GPT-5.4-mini on low-overlap apparatus questions only. On public-domain real-world questions, no model shows a $\geq 5\%$ Brier improvement, and the per-source direction within “public-domain” depends on the question category. Pre-registration prevented us from over-stating the cross-corpus claim as a general failure-of-transfer; the specific finding is the corpus-dependent transfer.

10 Refined mechanism for the worry-scalar sign-flip: shared topic-trigger $\times$ per-family accuracy

A reviewer-flagged concern is that the affect-proxy hypothesis (§5 early framing) was still just a hypothesis. We ran four mechanism probes on existing within-program data to disambiguate.

Probe 1. Within-family rank correlation between (across-family p -variance per contract, used as a proxy for intrinsic question difficulty) and (that family’s mean worry on that contract). GPT-5.4-mini shows $\rho = -0.43$, 95% CI $[-0.69, -0.07]$, **h1_supported negative**, the opposite direction from a “question-difficulty confound” mediator. The mediator hypothesis is rejected.

Probe 2. Stratified $\rho(\text{worry}, \text{Brier})$ by question difficulty (hard half vs easy half) and by point-estimate extremity (uncertain $p \in [0.4, 0.6]$ vs confident). Claude’s inversion lives in the EASY-contracts subset only ($\rho_{\text{EASY}} = -0.19$, $\rho_{\text{HARD}} = +0.03$); GPT-5.5’s inversion lives in the UNCERTAIN- p subset only ($\rho = -0.31$, $n=21$). The sign-flip is family-specific AND subspace-specific; no single uniform mechanism.

Probe 3. Lexical log-likelihood-ratio of distinguishing tokens between high-worry ($\text{tip} \geq 70$) and low-worry ($\text{tip} \leq 30$) rationales per family. Under affect-proxy, high-worry rationales should be enriched in CAREFUL (*analyze, consider*) or UNCERTAIN (*unsure, difficult*) words. We observe instead that the dominant distinguishing terms are *specific content-topic words, tactic, convolution*,

measurability, singularity, consistent with a topic-trigger rather than affect-proxy hypothesis.

Probe 4. Cross-family worry agreement: do families AGREE on which contracts get high worry? Pairwise Spearman correlation between each pair of families’ mean-worry-per-contract gives Claude $\leftrightarrow$ GPT-5.5 $\rho=+0.40$ (**h1_supported**), GPT-5.5 $\leftrightarrow$ GPT-5.4-mini $\rho=+0.41$ (**h1_supported**), GPT-5.4-mini $\leftrightarrow$ DeepSeek $\rho=+0.64$ (**h1_supported**). **Families share a topic-trigger signal: they pick the same contracts as high-worry.**

Probe 5. $\rho(\text{worry}, \text{wallclock_seconds})$ per family, does the model “think longer” on high-worry rows? All families show $|\rho| \leq 0.25$ with CIs crossing zero. Worry is NOT a compute-effort proxy.

Probe 6, mean-regression within row. For each row we measure $\rho(\text{worry}, |p_{\text{success}} - 0.5|)$ per family. Claude: $\rho = -0.39$ 95% CI $[-0.55, -0.20]$, **h1_supported** negative. Codex-mini: -0.59 $[-0.72, -0.41]$, **h1_supported**. DeepSeek: -0.35 $[-0.58, -0.06]$, **h1_supported**. GPT-5.5 and Gemini both null. **Three of five families moderate their probability when emitting high worry.**

Probe 7, per-contract accuracy alignment. Rank contracts by claude’s mean worry. Compare mean Brier per family in the top-quartile (claude-high-worry contracts) vs bottom-quartile (claude-low-worry contracts):

Family	Top-quartile Brier	Bottom-quartile Brier	Diff
Claude Opus 4.7	0.181	0.392	−0.211
GPT-5.5	0.199	0.224	−0.025
GPT-5.4-mini	0.160	0.219	−0.060

Claude is half as wrong on its high-worry contracts. The other families show small or no asymmetry.

Probe 8. RLHF hedging mediator (rejected). $\rho(\text{hedging-vocabulary density}, \text{Brier})$ is POSITIVE for Claude (+0.15) and GPT-5.4-mini (+0.31), opposite the direction of their worry-Brier inversion. Hedging-language is not the mediator.

Converged synthesis: a two-mechanism interaction. The worry-Brier sign-flip emerges from the simultaneous presence of two distinct mechanisms:

1. *Worry-triggered mean-regression.* High worry $\rightarrow$ less-extreme p on the same row (Probe 6). Three of five families show this at **h1_supported**.
2. *Per-family accuracy alignment with the trigger topics.* The shared topic-trigger (Probe 4) flags long-structured apparatus contracts as high-worry. Claude is GENUINELY BETTER on those contracts (Probe 7, diff -0.21); GPT-5.4-mini is not (diff -0.06).

Claude has BOTH (1) and (2): moderation lands close to truth on contracts where claude is accurate $\rightarrow$ inverted $\rho(\text{worry}, \text{Brier})$ (the sign-flip signature). GPT-5.4-mini has (1) but not (2): moderation doesn’t save it on topics it’s weaker on $\rightarrow$ direction-sensible $\rho(\text{worry}, \text{Brier})$. GPT-5.5 has neither cleanly at the pooled-across-conditions level, and its worry-Brier inversion (documented at the per-pilot positive-magnitude-balance-off cell, $N=85$) is condition-specific.

The mechanism is identified at evidence-summary grade for Claude (the most cleanly-inverted family). It is partial for the other inverted families. White-box activation-level probing would directly test which activations drive worry emission versus probability emission; we have only the black-box behavioral signature. The worry scalar by itself is not a competent calibration signal across families. The diagnostic interpretation is now sharper: per-(family, corpus) analysis is needed because mean-regression is universal-but-variable, while per-family topic-accuracy alignment is what gives the worry-Brier relationship its sign.

11 Multi-channel R^2 and a NEW channel: self-predicted Brier interval

The orthogonality result (§8) said the three channels are statistically independent. A natural follow-up: how much of forecast Brier do they jointly explain when measured *on the same forecast*? Earlier we could only estimate single-channel R^2 per primitive (5–26%) because each primitive was elicited in a separate pilot with its own forecast. The all-channels primitive elicits four channels in ONE prompt: bid-ask spread, scalar worry, AND the lower and upper bounds of the LLM’s self-predicted Brier interval (a previously-unused continuous channel).

Multi-channel OLS R^2 on same-forecast Brier (public-domain $N=42$, leave-one-out cross-validated):

The reported R^2 on $N=15$ low-overlap with $k=4$ channels is mathematically reckless, 14 degrees of freedom and four free regressors gives an in-sample R^2 that is mostly fit noise. The honest report is the public-domain $N=42$ corpus with adjusted R^2 and leave-one-out R^2 (LOO), where LOO subtracts the optimism bias:

Family	N	R^2 (in-sample)	adj R^2	R^2 LOO	reading
GPT-5.5	42	0.572	0.525	+0.312	real out-of-sample signal
Gemini 2.5 Flash	42	0.305	0.230	+0.042	barely real
Claude Opus 4.7	42	0.201	0.115	+0.027	barely real
DeepSeek Chat	42	0.276	0.197	−0.053	overfit
GPT-5.4-mini	42	0.162	0.072	−0.288	overfit worse than the mean

The honest claims.

First, *only GPT-5.5 has a multi-channel decomposition that survives leave-one-out cross-validation on the initial public corpus*. For GPT-5.5, four channels (worry scalar, bid-ask spread, b_mid, b_width) jointly explain $\approx 31\%$ of squared-error variance out-of-sample at $N=42$. For the other four families, R^2 LOO is at or below zero, the multi-channel decomposition does not generalize beyond the fit set at this N , even though three of them have non-trivial in-sample R^2 . The orthogonality claim (§8) stands; what does NOT stand is the stronger claim that combining the channels reliably predicts error for those four families at $N=42$.

A later no-call re-audit on the larger v28a pool ($N=157$ per family; 785 all-channel rows) further scopes the result. Using only the four emitted channels, all five families have negative leave-one-out R^2 for Brier prediction (Claude −0.059, GPT-5.4-mini −0.050, GPT-5.5 −0.065, DeepSeek −0.024, Gemini −0.016). Adding those channels to the shortcut baseline `question_len + p.success` yields negative incremental LOO R^2 for all five families. Some Polymarket sub-source cells show positive channel-only R^2 , but the shortcut model is usually as strong or stronger. The channel result is therefore an elicitation-surface diagnostic, not a standalone classifier.

Second, *a new continuous channel was introduced and is properly characterized as a primitive, not a fix*. The all-channels prompt asked each model to emit `predicted_brier_lo` and `predicted_brier_hi` alongside the forecast. The interval midpoint (b_mid) and width (b_width) form a self-predicted-Brier channel with no precedent we are aware of in the LLM-calibration literature. On the public-domain corpus, b_mid and b_width each carry $\rho < 0.30$ versus Brier for every family, they are informative for one family (GPT-5.5 multi-channel survivor) but are not a single-channel rescue.

What the low-overlap $N=15$ numbers were. A within-program control on research-internal contracts, with $k=4$ regressors at $N=15$, the in-sample R^2 of 0.66 to 0.90 values for GPT-5.4-mini

and Gemini are in-sample fits with too few residual degrees of freedom to interpret. We report them in our internal logs; we do NOT make a cross-corpus deployment claim from them.

Partial replication of the reference-class outside view at $N=42$ public-domain. A separately-fired pilot elicited an outside-view base rate (the model first listed five similar past resolved questions and emitted their historical YES rate as `p_base_outside`, before its own `p_success`). At $N=42$ public-domain, $\rho(\text{p_base_outside}, y)$ is positive in direction for all five families and reaches $\rho=+0.39$, 95% CI $[+0.10, +0.62]$, `h1_supported` for GPT-5.4-mini, the only family that clears the Fisher-z detectability floor at $N=42$. For that family the outside-view tracked outcomes more strongly than its own point estimate ($\rho=+0.21$). The other four families are `inconclusive_underpowered`, all positive-direction. We treat this as a partial cross-corpus replication of a novel primitive, not a full generalization.

Saturation curve. The R^2 -vs-channel-count saturation curve is the right summary plot for policy translation; we leave it pre-registered as the next analytical pass, on the public-domain corpus and reported in LOO terms.

12 Where the per-family signal actually lives: conditional structure

The §11 multi-channel result (only GPT-5.5 survives LOO on $N=42$) leaves four families with channels that don’t generalize. Three free post-hoc analyses on the same data sharpen this:

Sub-source heterogeneity within the public-domain corpus. The $N=42$ public-domain corpus is itself a mixture of Polymarket prediction-market questions, Manifold prediction-market questions, and yfinance stock-threshold questions. Recomputing $\rho(\text{worry}, \text{Brier})$ per (family, sub-source):

Family	yfinance ($N=10$)	Manifold ($N=19$)	Polymarket ($N=13$)
Claude	+0.54	+0.51 (h1)	−0.44
GPT-5.5	+0.59	+0.20	+0.64 (h1)
GPT-5.4-mini	−0.69 (h1)	−0.12	−0.38
Gemini	+0.13	−0.21	+0.75 (h1)
DeepSeek	+0.21	−0.28	+0.95 (h1)

Claude’s sign flips between Manifold (+0.51) and Polymarket (−0.44); DeepSeek’s signal lives almost entirely in Polymarket (+0.95, 95% CI $[+0.82, +0.98]$). Pooled-corpus ρ averages across this heterogeneity. The per-family rule is incomplete without a sub-source decomposition, a contract’s host platform is a hidden confound.

Pooled signal collapses after task-difficulty controls. A standard practice is to report “classifier on LLM uncertainty channels predicts $R^2 \approx 0.36$ of Brier variance.” We refit on the pooled $N=210$ (5 families $\times$ 42 contracts) with two task-difficulty baselines added: `question_length` and forecast-extremity $|p_{\text{success}} - 0.5|$. Incremental R^2_{LOO} from each LLM channel added on top of the baseline:

- baseline (q_len + p_extremity only): $R^2_{\text{LOO}} = +0.101$
- + worry: $\Delta = -0.003$
- + spread: $\Delta = +0.011$
- + b_mid: $\Delta = -0.009$

- + b_width: $\Delta = -0.010$

On pooled data, controlling for task-difficulty wipes out the LLM-channel contribution. The per-family multi-channel signal (GPT-5.5 $R^2_{\text{LOO}} = +0.31$, §11) survives because it conditions on family identity. A claim of the form “LLM-emitted uncertainty channels add predictive value” *across* families on pooled data does NOT hold once task-difficulty is controlled.

Hardness-conditional multi-channel R^2 . Per-(family, hardness-quartile) cell on $N=42$ public, where hardness = cross-family $\sigma(p_{\text{success}})$ (high σ = families disagree = hard):

Family	Q1 (easy, $N=11$) R^2_{LOO}	Q4 (hard, $N=10$) R^2_{LOO}	Reading
GPT-5.5	+0.93	−0.79	generalizes on easy; overfits on hard
DeepSeek	+0.92	−13.0	same, near-singular on hard
Gemini	+0.87	−2.5	same
GPT-5.4-mini	−0.74	−2.0	no fit either bucket
Claude	−4.1	−10.0	no fit either bucket

On the easy quartile (low cross-family disagreement, $N=11$), three families show multi-channel $R^2_{\text{LOO}} > 0.87$. On the hard quartile (high disagreement), LOO collapses across all families. The cell size is small ($N=10-11$ with $k=4$ channels), so this is a hypothesis-generating result requiring $N \geq 20$ per cell to confirm; the deepseek Q3 $R^2_{\text{LOO}} = -480$ is a near-singular-design artifact. The pre-registered confirmation pass is on the $N=142$ topic-balanced extension.

The policy hypothesis from these three findings is one scoped rule: *test per-family multi-channel routing only when sub-source identity is matched and cross-family disagreement is low*. Later re-audits demote current channel-only routing to diagnostic status, so the claim here is about where an uncertainty signal appears, not a validated deployment rule. Stated differently, the LLM-uncertainty signal is conditional on context-of-call, not unconditional per-family.

Cross-corpus-class Brier and Elo: the family ordering flips between internal apparatus and external markets. A per-(family, corpus_class) roll-up across all 15,949 panel calls (post-ingest 2026-05-29; SQL views `v_family_brier_by_corpus_class` and table `family_elo_by_corpus_class` in `forecaster_calibration.db`). Elo: $K=16$, init 1500, head-to-head per-contract Brier wins on the shared cohort.

Family	Corpus class	n calls	Brier	Elo	n games
GPT-5.4-mini	external	1291	0.279	1479	568
GPT-5.5	external	1285	0.282	1463	568
Claude	external	1429	0.285	1575	568
DeepSeek	external	1396	0.295	1464	568
Gemini	external	1427	0.327	1519	568
GPT-5.4-mini	internal	516	0.135	1681	118
GPT-5.5	internal	573	0.208	1416	119
DeepSeek	internal	342	0.226	1526	60
Gemini	internal	342	0.236	1569	60
Claude	internal	580	0.248	1308	119

Two readings the table forces. First, Brier and Elo orderings disagree externally: Claude wins external Elo at 1575 but is 3rd on external Brier (0.285); Gemini is 2nd on external Elo at 1519 but 5th on external Brier (0.327). The disagreement is mechanistically informative: a family can win head-to-head by being direction-correct more often than its calibration confidence suggests, while

losing average Brier on the few contracts where it bets confidently and wrong. Gemini’s external $\Delta = 56$ Elo above its Brier rank corresponds to roughly 58% head-to-head win rate against the next-down family despite its higher mean squared error. Second, the family ordering flips between corpus classes. Claude moves from external Brier 0.285 / Elo 1575 (best external Elo) to internal Brier 0.248 / Elo 1308 (373 Elo points below GPT-5.4-mini, $\approx 89\%$ head-to-head win rate against Claude on internal). The mechanism is the F100 confident-NO bias surfacing on the internal slate: per-family mean p_{success} on internal contracts is 0.345 for Claude versus the true internal YES rate of 0.556, a systematic under-bet on the 56% of internal contracts that resolve YES; GPT-5.4-mini’s mean internal $p_{\text{success}} = 0.556$ matches the base rate exactly, with 33% extreme bets ($p \geq 0.9$ or $p \leq 0.1$) that are well-calibrated rather than conservative. Internal N at resolved-contract level is 45, so the largest per-family Elo gap (GPT-5.4-mini vs Claude $\Delta=373$) is `h1_supported` and tighter pairs are `inconclusive_underpowered`; an internal-corpus re-fire to $N \geq 142$ resolved parity is queued. The candidate policy shape is conditional routing by Elo where Brier least discriminates, keyed on both contract-level panel σ and corpus class; later source-balanced policy audits keep this below deployment status.

Ensemble feasibility probe (apples-to-apples on the $N=157$ shared cohort). A direct test of whether naive aggregation can substitute for the per-family routing analysis above: on the 157 contracts where all 5 families fired, the best single family (Claude, Brier 0.2528) beats both mean-of-5 (Brier 0.2558, $\Delta=+0.0030$) and median-of-5 (Brier 0.2704, $\Delta=+0.0176$). Naive aggregation does NOT beat the strongest single at this N . The reason is in the panel’s correlation structure: pairwise Spearman $\rho(p_{\text{success}})$ across families splits the panel into two clusters, (Claude, GPT-5.5, GPT-5.4-mini) within-cluster $\rho \approx 0.74\text{--}0.78$, (DeepSeek, Gemini) within-cluster $\rho = 0.66$, between-cluster $\rho = 0.40\text{--}0.60$. The Claude+codex trio is highly correlated; real diversity comes from Gemini and DeepSeek but they have higher individual Brier. A second ordering, head-to-head per-contract Elo ($K=16$, init 1500), surfaces a different signal: Claude 1555, *Gemini* 1516, GPT-5.4-mini 1512, GPT-5.5 1464, DeepSeek 1453. Gemini gets DIRECTION right more often than its calibration suggests; a family can win Elo head-to-head (closer to truth on more contracts) while losing average Brier (worse on the contracts it loses, especially with overconfident direction-correct extreme p_{success} values getting punished hard on the few it gets wrong). The ANOVA-style variance decomposition completes the picture: 67.1% of Brier variance is between-contract (some contracts are just harder), 32.3% is family $\times$ contract interaction (*different families err on different contracts — the actual ensemble opportunity*), and only 0.6% is between-family identity. Naive mean and median cannot capture the 32% interaction because they assume independent family errors of similar magnitude; the policy hypothesis that would convert this signal is contract-level conditional routing keyed on (panel disagreement σ , contract category, family Elo on similar- σ contracts). A pre-registered $N \geq 300$ confirmation pass is required before treating that router as a deployment rule.

13 Five additional findings from the all-channels primitive round

The all-channels round elicited multiple new primitives in a single prompt or as paired conditions. Five additional findings emerge at $N=131\text{--}142$ public-domain:

Rollback hurts. A self-counterfactual primitive (“imagine you were trained only up to 2023, emit a rollback forecast”) HURTS Brier for 4 of 5 families at $N=142$ paired (Δ Brier rollback–normal: claude +0.064, codex-5.5 +0.034, codex-5.4-mini +0.028, deepseek +0.028, gemini –0.005). Post-2023 training is net-useful even though families can’t introspect WHICH contracts benefit. *Operational caution:* don’t prompt-engineer “imagine you knew less” framings on forecasting tasks.

Leakage agreement without calibration. On the same rollback primitive, all 10 pairwise $\rho(|\text{leak}_A|, |\text{leak}_B|)$ between families are $\geq +0.43$ (h1_supported; claude $\leftrightarrow$ codex-5.5 $\rho = +0.80$). Families AGREE which contracts feel high-leakage. *But* per-family $\rho(|\text{leak}|, \text{err}^2)$ is h0_kept for all five at the ± 0.30 target. Self-reported leakage is a real CONTRACT-structural signal, NOT a per-family calibration channel. Useful as a corpus-curation filter (filter to low-leakage contracts for honest LLM-fresh tests).

DeepSeek’s emitted decision-threshold is calibrated. An asymmetric-loss decision-threshold primitive (commit-YES threshold under symmetric / YES-bad-3 $\times$ / NO-bad-3 $\times$ loss). Codex-5.4-mini emits the most Bayes-coherent thresholds (mean YES-bad 0.750, exact match). Deepseek h1: $\rho(\text{threshold_yes_bad}, \text{err}^2) = -0.19$, 95% CI $[-0.35, -0.03]$ at $N=142$. The emitted threshold tracks accuracy for one family. This is a novel diagnostic channel; policy use requires a heldout utility test.

Stake-framing amplifies worry universally, leaves p unchanged. Same contract framed as “\$1 stake” vs “\$100K stake” across 103–142 paired contracts per family. Worry-scalar amplifies for every family (Δworry from +10.8 deepseek to +33.5 claude on the 1–100 scale). But $|\Delta p_{\text{success}}| < 0.03$ for every family. Universal cognitive-shift-without-calibration-shift on a new manipulation, bridges the cognitive-decoupling result to the stake-framing / risk-amplitude human literature (Holt-Laury 2002).

Inversion rule for channel-routing meta-cognition (three of five families). A meta-channel-routing primitive (“which of your own uncertainty channels best predicts your error?”) reveals systematic anti-coherence at $N=15$ internal: claude, codex-5.5, and deepseek pick the WRONG channel 87% of the time (binomial $p=0.005$ each). Within those picks, $\rho(\text{picked}, \text{err}^2)$ is negative while $\rho(\text{unpicked}, \text{err}^2)$ is strongly positive (claude +0.51, codex-5.5 +0.53, deepseek +0.65). The candidate inversion rule is: when these three families claim “worry is my best channel,” test spread instead (or the analogous flip). Codex-5.4-mini is coherent (73% right); gemini is random. Companion finding to the per-family worry sign-flip, on a new axis. Internal-only at $N=15$; cross-corpus replication at $N=42$ external is pre-registered. The pattern (LLMs have honest contract-level structural intuitions without calibration to truth) recurs across the rollback, leakage, decision-threshold, and meta-channel-routing results, the defining negative finding about LLM self-monitoring.

14 Universal cross-family patterns

The findings above split the LLM panel by family. Free analyses on the same $N=42$ public-domain data surface four universal patterns that hold across all five families with the same direction, which we report here because they are the patterns most likely to transfer to a different LLM panel.

Universal over-confidence on confident-NO predictions. A reliability-curve analysis (binning each family’s forecasts into quintiles by stated p_{success} , then computing the actual YES rate per quintile):

Family	Mean predicted p in quintile 1	Actual YES rate, quintile 1	gap
Claude	0.04	0.38	+0.34
GPT-5.5	0.07	0.75	+0.68
GPT-5.4-mini	0.04	0.62	+0.58
Gemini	0.05	0.75	+0.70
DeepSeek	0.03	0.75	+0.72

Every family is dramatically over-confident on its “confident-NO” forecasts. The quintile-5

(confident-YES) end is roughly calibrated (gap -0.02 to -0.21). The miscalibration is asymmetric: confident-NO is poorly calibrated; confident-YES is OK. With $N=8-10$ per quintile per family, the per-family CI is wide, but the universality across all five families and the consistent direction of the gap argue the pattern is real. *Operational caution:* on a news-event public corpus, do not trust LLM $p_{\text{success}} < 0.10$ as a confident-NO commit without heldout correction.

Horizon-conditional Brier. Pooled across all 5 families ($N=210$): $\rho(\text{days to resolution from 2026-01-01, err}^2)$ $+0.161$, 95% CI $[+0.026, +0.290]$, h1_supported. Brier on April-2026-resolving contracts is 0.240; on May-2026 it is 0.385. Contracts farther from the training cutoff are systematically harder. The mechanistic interpretation is straightforward (more uncertainty accumulation over longer horizons), but the cleanly-documented per-contract horizon-confidence-weighting term is novel for LLM forecasting at this N .

Universal NO-bias. On contracts that did resolve YES ($N=26$ of 42 in this corpus), every family’s mean p_{success} is below 0.5 (0.38–0.48). Brier conditioned on $y=1$ exceeds Brier conditioned on $y=0$ for every family; DeepSeek shows the largest asymmetry (0.46 vs 0.12). LLMs are systematically better at predicting NO outcomes than YES outcomes on this corpus. Possible interpretations include (a) underlying corpus YES base-rate of 62% exceeds the model’s prior on these questions; (b) “will X happen by date Y” framing biases toward NO; (c) prior-cutoff bias on then-future events. We do not distinguish among these here.

Naive ensemble averaging does NOT help. On the full $N=142$ paired set with all five families’ p_{success} values: median-of-5 Brier = 0.272, mean-of-5 Brier = 0.261, best-single-family (Claude) Brier = 0.254. Paired permutation: median-of-5 loses to the best single family ($\Delta=+0.018$); mean-of-5 also loses ($\Delta=+0.007$). The standard “ask five LLMs, take the median” aggregation strategy yields no benefit on this corpus and slightly hurts. Best-single-family dominates naive ensembling at this N .

Universal cross-family forecast agreement. For every pair of the five families, $\rho(p_{\text{success},A}, p_{\text{success},B})$ on the same $N=42$ public-domain contracts is h1_supported. Pairwise ρ ranges $+0.52$ (codex-5.4-mini $\leftrightarrow$ DeepSeek) to $+0.88$ (Claude $\leftrightarrow$ GPT-5.5). Mean $+0.72$. The five families agree on forecast direction across vendors and architectures. This caveats the per-family results: when families converge, they really converge, but high pairwise ρ does not distinguish “models converge on truth” from “models share training-data bias.” For panel-aggregation design, the practical takeaway is that consensus-strength is a usable confidence weight: a contract where all five families return near-identical p_{success} can be treated as a higher-confidence pooled forecast than a contract where they disagree.

Universal source-difficulty ordering. Per-source Brier per family on the public-domain corpus reveals a universal ordering: *Polymarket* $>$ *Manifold* $>$ *yfinance*. Polymarket contracts (election outcomes, news-events) yield Brier 0.46–0.57 across all five families; yfinance ETF threshold contracts yield 0.22–0.23. The same order holds for every family. Source identity is a per-contract confidence-weighting term independent of family.

The six universal patterns together suggest a candidate per-contract recipe: (a) discount the model’s confident-NO forecasts; (b) horizon-weight per-contract confidence; (c) apply a per-family YES-bias correction term; (d) prefer a calibrated single family over a naive ensemble; (e) weight pooled-forecast confidence by cross-family consensus-strength; (f) weight by source difficulty.

Composed recipe at $N=142$: routing beats naive aggregation; later audits demote the composite policy. We instantiate the rules above as a single composed forecast (the mean-of-5 panel forecast adjusted by four universal coefficients: a confident-NO discount when the panel mean is below 0.20, an upward YES-bias correction in the middle band, a horizon-to-cutoff shrinkage, and a per-source-difficulty shrinkage, see App. A for the closed form). All four coefficients are chosen *ex ante* from the universal patterns above, not learned from held-out folds. The contrast we

test is paired-permutation Δ -Brier against three baselines and against a per-family channel-routed alternative on the same $N=142$ paired contracts.

strategy	Brier	Δ vs. best-single	90% CI	p_{perm}
median-of-5	0.272	+0.018	,	,
mean-of-5	0.261	+0.007	,	,
best-single (Claude)	0.254	,	,	,
routed (universal-only)	0.232	-0.022	$[-0.050, +0.005]$	0.18
routed (per-channel)	0.246	-0.009	$[-0.035, +0.016]$	0.58

Three results.

(i) *Routed-vs-naive-aggregation is the clean within-cohort win.* Routed (universal-only) beats median-of-5 by $\Delta=-0.040$ ($p_{\text{perm}}=0.0013$) and mean-of-5 by $\Delta=-0.029$ ($p_{\text{perm}}=0.0069$). **h1_supported**. The composed recipe out-performs the standard “ask five LLMs, take the median/mean” baseline at $p<0.01$ on the same $N=142$ paired set, but this comparison is weaker than a source-balanced policy audit against the strongest simple correction.

(ii) *Routed-vs-best-single is directional but underpowered.* The point estimate $\Delta=-0.022$ sits almost entirely on the routing-wins side of zero (90% CI $[-0.050, +0.005]$), and the direction is consistent with the same comparison at the previous $N=42\times 3$ snapshot before the codex refill ($\Delta=-0.013$, $p_{\text{perm}}=0.48$). The gap roughly doubled in routing’s favor as N rose; the p shrank from 0.48 to 0.18. At the observed effect size and paired-Brier variance, the Fisher- z power calculation gives $N\approx 250-300$ for $p<0.05$ at 80% power on a two-sided test. We report **h0_kept** against the $\Delta\geq 0.05$ detection bar that the program’s pre-registration sets for deployment claims, and **inconclusive_underpowered** against the smaller observed effect.

(iii) *Universal rules generalize better than per-family channel routing in this audit, but the composite does not survive as the live policy.* Routed (universal-only) beats routed (per-channel) by $\Delta=+0.014$ ($p_{\text{perm}}=0.042$), i.e. the per-family channel-weighting alternative makes the forecast *worse* than the universal-rules version on the public-domain corpus, at $p<0.05$. This is consistent with §8 and with the later channel-only re-audit: channel-level routing is contextually conditional and does not survive a uniform application. Later source-balanced policy audits further show that the current source+ σ router and diagnostic-triggered allocation lose to simpler rules. A costed review-allocation audit shows oracle headroom but does not promote any realistic proxy reviewer. A no-call expert-advice continuation over the same complete-five panel cohort gives the same applied verdict: Hedge over raw-family, F100-adjusted-family, and simple-pool experts weakly beats confident-NO mean-panel overall (0.226481 versus 0.233529; $\Delta = -0.0070$, $p = 0.4671$) but fails promotion and regresses on Manifold in the balanced Manifold/Polymarket slice. The oracle-expert Brier remains far lower (0.117454), so family-choice headroom is real but not yet recoverable by the current observable policies. One prompt-geometry exception survives in bounded form: a source-balanced four-family F47 contrastive packet promotes pairwise ranking over 24 unique non-tie pairs (unique accuracy 0.750, utility +0.583, $p = 0.0044$ vs random). A follow-up overlapping four-family tournament then promotes a source-heldout pairwise-to-probability translation against the prompt’s own probabilities (raw Brier 0.234719, translated 0.200417, $\Delta = -0.034302$, $p = 0.0050$, no source regression). A stricter same-packet policy control is directionally favorable but below the production gate: translated panel Brier is 0.178411 versus raw panel 0.198773 and mean-family F100 0.201126, but translated-minus-F100 has $\Delta = -0.022714$ with $p = 0.0628$, and same-contract market overlap is only three rows. A later cross-packet transfer audit strengthens but does not close the case: training the translation on the source-balanced packet and testing on the tournament packet gives translated panel Brier 0.175687 versus mean-family F100 0.201126 ($\Delta = -0.025439$,

$p = 0.0314$), while the reverse transfer remains favorable but misses the panel gate (0.171162 versus 0.196125, $\Delta = -0.024963$, $p = 0.0636$). The first external-bar control is a warning rather than a promotion: after public Manifold acquisition and corrected Polymarket direct-slug plus intraday-fidelity acquisition, the joined slice is still only twenty-four rows. Translated F47 has Brier 0.169991, the 50/50 market+F47 blend 0.170067, raw panel 0.176079, market-alone 0.183011, and mean-family F100 0.185751; translated-vs-market has paired $p = 0.5783$ and translated-vs-raw has $p = 0.7351$. A prospective Polymarket packet has now frozen 24 pairwise market bars before any model calls, but it is a queue for future resolved scoring rather than outcome evidence. This is an experimental graph-derived translation layer, not generic self-repair and not yet a market/human comparison. The surviving production point-probability rule remains the confident-NO discount until F47 translation clears prospective or market/human-joined F100/raw controls.

A ~ 100 -contract pre-registered follow-on on the public-domain corpus was the original refill required to escalate (ii) from **inconclusive_underpowered**. The newer evidence changes the applied question: the next policy test must beat the confident-NO discount and source/hash controls, not only naive mean or median aggregation.

The confident-NO discount as a standalone rule beats raw on every family at $p < 0.05$. The simpler form of the recipe, the single confident-NO discount applied per-family without the other three rules, improves per-family Brier on every panel member at $p < 0.05$:

family	raw Brier	discounted Brier	Δ	paired p_{perm}
claude	0.2543	0.2240	-0.0302	0.016
codex-5.5	0.2625	0.2416	-0.0208	0.030
codex-5.4-mini	0.2714	0.2450	-0.0264	0.015
deepseek	0.3222	0.2704	-0.0518	0.0008
gemini	0.3167	0.2840	-0.0327	0.008

Two observations land here. (a) Discounted Claude at Brier 0.224 is at parity-or-better than the four-rule routed recipe (0.232 above); the additional three rules add nothing meaningful on top of the discount on this corpus at $N=142$. (b) A single one-line post-forecast adjustment improves Brier at $p < 0.05$ across every model family, including outside the panel that originated the rule (DeepSeek and Gemini are subscription-distinct families). The current applied shape is the discount alone, applied to whichever single model is in production and tracked against source-balanced controls. A later no-call fitted-calibrator audit did not promote a replacement: source-isotonic slightly improved the overall point estimate but lost Manifold and was non-significant, while tail-beta shrinkage was worse than the hand F100 rule. The 2026-06-04 policy-scoreable rerun excludes 10 yfinance/yfinance.etf complete panels lacking label-time receipts, leaving 132 panels; source-isotonic remains non-promoted at -0.005248 Brier versus confident-NO with paired $p = 0.7099$, while raw mean-panel is $+0.029598$ worse. A later source-currency stress audit on the Law 3 Stage-B panel narrowed this rule: it improved post-cutoff rows (Brier delta -0.025326 ; tail-only -0.101306) but regressed pre-cutoff/source-visible rows (delta $+0.035016$, $p = 0.0002$; tail-only $+0.097719$, $p = 0.0002$). A receipt-level rerun through the shared source-currency discriminator left those scores unchanged while exposing 39/240 stored-flag-vs-computed-relation conflicts, so the rule is now explicitly tied to computed cutoff and label-time receipts; the same discriminator is materialized in a call-level DB gate for downstream cutoff-stratified audits. We therefore treat the discount as forward-looking calibration, not as retrospective benchmark correction.

15 Bias-type-specific inheritance: LLMs escape utility-grounded biases and inherit framing-grounded biases

Three novel-bias smokes on the same $N=30$ public-domain contracts across all five families ($5 \times 5 \times 30 = 750$ calls, 748 schema-OK) test whether well-documented human cognitive biases transfer to LLM forecasting. The findings split cleanly along bias TYPE:

(A) Loss-frame invariance. LLMs ESCAPE the human bias. The same contract asked as “ p_{success} ” vs “ p_{failure} ” should satisfy $p_{\text{success}} \approx 1 - p_{\text{failure}}$ for a symmetric forecaster; humans show systematic gaps of 0.15–0.30 from loss aversion (Kahneman–Tversky 1979). Mean per-family $|p_{\text{success}} - (1 - p_{\text{failure}})|$ across all five panel members: claude 0.035, deepseek 0.049, codex-5.5 0.085, codex-mini 0.098, gemini 0.148. All five families come in well below the human reference, with claude and deepseek essentially symmetric (median ≤ 0.03).

(F) Status-quo / endowment effect. LLMs INHERIT the human bias, at large effect size. The same contract framed as “the YES condition CURRENTLY HOLDS at the time of forecast” vs “do not assume any current state” should satisfy $p_{\text{currently}} \approx p_{\text{fresh}}$ if the model is anchor-symmetric; humans show endowment-effect anchoring on stated state. Mean per-family $|p_{\text{currently}} - p_{\text{fresh}}|$: gemini 0.084, deepseek 0.110, claude 0.241, codex-mini 0.275, codex-5.5 **0.440** (median 0.500, half of contracts flip sign between the two framings). Three of five families show a clearly large status-quo response; codex-5.5 in particular reverses the forecast on half the contracts purely from prompt framing.

(B) Probability-weighting curve, preliminary near-linear (no inverse-S). For each contract we elicit p_{emit} at the eight anchor probabilities $\{0.01, 0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.99\}$. Cumulative prospect theory (Tversky–Kahneman 1992) predicts inverse-S overweighting of small probabilities. Preliminary Claude responses: $\{0.02, 0.07, 0.12, 0.27, 0.50, 0.73, 0.88, 0.985\}$. Near-linear; no CPT inverse-S overweighting at the low-probability tail.

The split is informative. The biases that DO transfer (status-quo / endowment) are grounded in stated state framing. LLMs anchor on what the prompt asserts. The biases that DO NOT transfer (loss-frame symmetry, probability-weighting curve) are grounded in human utility structure. LLMs have no value function over loss vs gain, no consumption preference, no asymmetric disutility of loss. This is the clean diagnostic shape: *LLMs inherit framing-grounded biases and escape utility-grounded biases*. It is neither the “LLMs are human-like” nor the “LLMs are rational” framing the literature has been working with.

The candidate rule the split suggests: when contract framing is ambiguous on current-state, elicit BOTH framings and average. Two-arm cost ($+2\times$) on ambiguous contracts, near-zero cost on unambiguous ones, eliminates the codex-5.5 half-flip pattern in (F). An $N=42$ external replication on a diversified public-domain corpus (Metaculus + FRED, 90 contracts, post-cutoff filtered) is firing at the time of writing.

16 Prompt-invariance asymmetry across families: a methodological caveat

Across three prompt variants that elicit p_{success} on the same public-domain contracts (all-channels, reference-class outside view, and conjunctive decomposition), per-family cross-prompt stability is heterogeneous:

Mean per-family standard deviation of p_{success} across the three prompts is 0.05–0.06 for Claude/Codex variants; 0.108 for Gemini; 0.148 for DeepSeek. Claude and the two Codex variants emit beliefs that are 2–3 $\times$ more prompt-invariant than Gemini and DeepSeek’s.

Family	$\rho(p_{\text{all-channels}}, p_{\text{outside-view}})$ at $N=42$	95% CI
Claude	+0.868	[+0.77, +0.93]
GPT-5.5	+0.830	[+0.70, +0.91]
GPT-5.4-mini	+0.870	[+0.77, +0.93]
Gemini	+0.504	[+0.24, +0.70]
DeepSeek	+0.386	[+0.09, +0.62]

This was initially read as the methodological caveat touching every per-family finding. We later refined it: pooling p_{success} across four “standard” prompts (all-channels, asymmetric-loss, rollback, low-stake) on the same ($N=41$ – 42) contracts per family shows mean $\sigma(p_{\text{success}}) = 0.042$ – 0.064 for EVERY family, including Gemini and DeepSeek. The earlier-reported $0.108 / 0.148$ values were driven specifically by the reference-class outside-view prompt, which forces a pre-forecast base-rate computation. That prompt is a different reasoning architecture, not the same forecast with prompt noise.

The scoped caveat is then: the per-family findings in this paper based on *standard* elicitation prompts (worry scalar, bid-ask spread, predicted-Brier interval) are roughly equally prompt-invariant across all five families on the POINT ESTIMATE p_{success} . Findings that require the model to execute a non-standard reasoning step pre-forecast (e.g., the outside-view computation) may shift more for Gemini/DeepSeek than for Claude/Codex.

Channel-level caveat: the worry scalar is roughly $4\times$ noisier across prompts than p_{success} . Pooling worry data from the all-channels, low-stake, and high-stake prompts on the same ($N=41$ – 42) shared contracts per family: mean per-family $\sigma(\text{worry})$ is 10 – 22 on the 1 – 100 scale (10 – 22% of full range), versus $\sigma(p_{\text{success}}) = 0.04$ – 0.06 (4 – 6% of $[0, 1]$). Stated equivalently: the worry channel carries substantial prompt-elicitation noise that the point estimate does not. This is a methodological caveat on every worry-based finding in this paper (the per-family sign-flip, the DeepSeek threshold result, the cross-prompt contradiction): the worry channel itself is less prompt-invariant than p_{success} . The clean disambiguation requires within-family worry-paraphrase pilot (5 prompt designs $\times$ same contract per family) at proper N ; we have not run it.

Cross-prompt sign-stability of the worry channel (critical). Pooling worry- p_{success} pairs from four public-domain prompt variants (all-channels, asymmetric-loss, rollback, stake-framing) at $N=248$ – 326 per family: $\rho(\text{worry}, \text{err}^2)$ for Claude is $+0.166$ [$+0.042, +0.285$], **h1_supported** in the DIRECTION-CORRECT direction, the opposite of the low-overlap-corpus result that drove the deployed per-family sign rule. Codex-5.5 is direction-correct externally (consistent with internal); Codex-5.4-mini’s strong internal direction-correct signal goes null externally; Gemini and DeepSeek remain flat. The cleanest read is that the inversion-for-Claude rule is corpus-bound or prompt-bound, not a per-family property. The deployed routing rule for Claude has been scope-restricted to “low-overlap-corpus prompt style only” pending a retest pilot (~ 42 calls of the original prompt style on $N=42$ public-domain) that disambiguates corpus-effect from prompt-effect.

17 Pre-registered test of the inheritance frame on a held-out bias slate

The three-axis frame in §3 was induced from the body of measurements above. The framework earns its predictive claim only by getting subsequent bias-transfer predictions right on biases it had not yet seen. This section reports the pre-registered test.

Pre-registered slate. Seven additional human biases held out from the F102 measurement, classified *ex ante* into the three cells of axis 2 before any new data was collected.

- INHERIT (pure-heuristic, high natural-speech footprint): **C** gambler’s / hot-hand fallacy [13]; **E** self-overconfidence [8]; **H** base-rate neglect when the prior is explicit; **J** numeric anchoring [14].
- ESCAPE (pure-motivational, low text representation of the bias mechanism): **G** hyperbolic discounting [2].
- MIMIC (systemic-motivational, intrinsic utility BUT heavy case-study representation): **D** sunk cost [1]; **I** in-group / national framing [10], with explicit prediction of alignment damping.

Decision rule. Of the seven held-out biases together with the three already measured in §15 (A loss-frame ESCAPE confirmed; B probability weighting ESCAPE preliminary; F status-quo MIMIC as re-classified from INHERIT), classifying ≥ 8 of 10 into the predicted cell is the bar for **h1_supported** on the framework as a structured prediction. The random-cell baseline is $\sim 3.3/10$. The MIMIC vs INHERIT discrimination has two candidate signatures: (S1) the magnitude correlates with how often the bias is *discussed* in training corpora (proxy: retrieval of the bias name returns case-study hits, not first-person enactments); (S2) the effect collapses under explicit anti-bias prompting much more for MIMIC than for INHERIT. S2 is reported as a failed/scoped companion test, not as part of the promoted law: the anti-bias-collapse score verdict is **kill_or_scope_raw_gap_explains_collapse**.

Slate verdict. On the pre-registered slate the framework classifies ≥ 8 of 10 biases into the predicted cell, clearing the **h1_supported** bar against the $\sim 3.3/10$ random-cell baseline. The two MIMIC biases pre-registered with explicit alignment damping—D sunk cost and I in-group/national framing—initially fell below the INHERIT magnitude band on the single-family smoke; on the cross-family panel both surface on the weaker-RLHF families at $3\text{--}6\times$ the Claude effect, exactly the pre-registered MIMIC-with-alignment-damping prediction, taking the slate to 10/10 on the per-family reading. We read this as confirmation that the transfer cell is predictable *ex ante* from each bias’s structural origin, with the alignment overlay (Axis 3) acting as a per-family modifier on the MIMIC class rather than a panel-uniform axis. Two extensions sharpen the claim beyond the categorical slate and are reported separately: (i) a quantitative dose-response relating transfer magnitude to each bias’s corpus-discussion frequency (signature S1), and (ii) a further out-of-distribution slate of biases drawn from outside the inducing set, classified *ex ante* under the same predicate. The pre-registration of cell classifications and decision rule is fixed at the version-tag of this manuscript.

Out-of-distribution slate (ii): inconclusive, and the apparent gradient is confounded.

We fired extension (ii) as a sealed pre-registration of nine fully-novel biases (none in the inducing set), scored as the *excess over a pre-registered normative framing gap* g_0 rather than the raw gap—so a model that gives the rational (framing-insensitive) answer reads as ESCAPE, never as a missed INHERIT. (An earlier instrument that scored raw gaps mislabelled rational answers as misses; that run is void.) On the five-family panel ($n=15$ events/bias, 1348/1350 schema-valid) the **full three-cell point-prediction is inconclusive**: median per-family cell-match is 5/9, below the pre-registered $\geq 7/9$ bar (random $\sim 3/9$). We also observed an ordering in which less RLHF-aligned families carry larger INHERIT/MIMIC excess (Claude lowest at +0.095; DeepSeek highest at +0.152; 3/4 adjacent pairs monotone) that would superficially support the Axis-3 alignment overlay. **We do not claim this as a result**, for three reasons. (1) It is underpowered: five families, a hand-assigned alignment ordinal, not a measured alignment quantity. (2) A cross-family contrast confounds alignment with

pretraining-corpus differences—families differ in pretraining as well as in RLHF, so a cross-family ordering cannot isolate the alignment stage. (3) The *direction* runs opposite to the established finding that instruction-tuning / RLHF *amplifies* several cognitive biases rather than damping them [4]. We therefore report slate (ii) as inconclusive on the cell-prediction and confounded on the alignment overlay. The framework’s validated contribution remains the in-distribution taxonomy and the MIMIC articulation; an out-of-distribution alignment law is *not* established here, and de-confounding alignment from pretraining (held-out within-family checkpoints, or matched-pretraining families) is the required follow-up.

18 The re-audit discipline: a warning to the field

The discipline that produced the results above also produced a re-audit of the program’s own prior findings. We treat the re-audit itself as a contribution because the same overcalling pattern almost certainly affects the published LLM-forecasting literature at the sample sizes it currently reports.

18.1 Power calculus applied to our own program

Before fixing the verdict resolver, eight prior “no effect” or “cross-corpus replication” claims in the program had been written down as findings. Applying the Fisher- z power-aware verdict resolver retrospectively:

- Four “null” findings move from `h0_kept` to `inconclusive_underpowered`: the data did not actively rule out a meaningful effect at the per-family target $|\rho|$ floor; the absence-of-signal was an absence-of-power.
- Five cross-corpus claims move from “replicated” or “failed-to-replicate” to corpus-specific: the original cross-corpus reading was a single sub-source point estimate at $N=5-15$ external, with CIs that crossed zero and with sub-source heterogeneity unmodeled.
- One self-knowledge-inversion finding (originally three families at $N=5$) retracts to a single family at $N=15$ internal; the other two cross-corpus directions flip on re-fire.
- Two per-family deployed routing rules (Brier- Δ map and *consider-both-sides* effect) move from deployment-grade to “directional hypothesis only”; the required N to confirm a $\Delta_{\text{Brier}}=0.05$ effect at 80% power is ≈ 459 per family, an order of magnitude above the original $N\approx 30$ per cell.

In aggregate, $\approx 50\%$ of the program’s prior verdicts did not survive the verdict resolver. None of the underlying executions were flawed. The interpretation discipline was.

18.2 A worked example: the per-family worry sign-flip retest

The previous YAML routing hypothesis for Gemini was scope-restricted to “low-overlap corpus, *v21*-style prompt only” pending a ~ 42 -call retest of the same prompt on the public-domain $N=42$ slice that the other *v28* pilots fired on. The retest, fired with the same C1 worry-only prompt on `gemini-2.5-flash` via API:

- Schema-OK: 42/42.
- $\rho(\text{worry, Brier}) = +0.110$ – the direction-correct direction, NOT the inverted direction the prior deployment had assumed.

- Fisher- z detectable $|\rho|$ at $N=42$, 80% power: ≥ 0.43 . Observed $|\rho|=0.11$ is well below this; the CI excludes the original “inverted” direction ($\rho < -0.30$).
- Verdict: `h0_kept` against the inversion hypothesis; the direction-correct point estimate is in the `inconclusive_underpowered` band for the positive hypothesis.

The retracted YAML rule for Gemini stays retracted. The earlier “inversion” claim, deployed at $N=5$ with $N_{\text{wrong}}=2$, was an artifact of the sample size. This is the single cheapest decisive retest the queue could fire; the queue carries two more retests of comparable diagnostic value.

18.3 Implications for the published LLM-forecasting literature

The two highest-profile 2024 contributions to LLM forecast calibration are (a) the ensemble-prediction result that a 12-LLM aggregate matches human-crowd Brier on real binary questions [9] and (b) the retrieval-augmented LLM that approaches human-superforecaster accuracy on a contamination-resistant slice [3]. Both report claims at sample sizes that, under the same Fisher- z power calculus we apply to our own program, are bounded by:

- For a Brier-delta $\Delta=0.05$ comparison (LLM ensemble vs. human crowd) at 80% power, paired permutation requires $N \geq 300$ –500 per side.
- For a per-model rank correlation $|\rho|=0.30$ at the same power: $N \geq 91$.
- For a per-(model, sub-source) cell at $|\rho|=0.40$: $N \geq 50$ per cell – requiring sub-source $N \geq 250$ across five families when sub-sources are five-way mixed.

We do not re-analyse those datasets here; we observe only that the standard “2024-style” aggregate claim is structurally compatible with the same overcalling failure mode our own program exhibited. The methodological contribution the published literature is missing – and that this paper contributes – is the verdict resolver discipline: every claim labelled with one of three legal verdicts, n_{required} pre-computed before fire, sub-source heterogeneity reported alongside the aggregate, and prior verdicts re-audited under the same calculus as new ones.

A second-order observation about contamination. The Halawi 2024 dataset (YuehHanChen/forecasting, $N=1754$ binary-resolved questions across Polymarket, Metaculus, Manifold, GJOpen, and CSET) reports resolution-date histogram: 1 question in 2021, 147 in 2022, 1470 in 2023, 136 in 2024, and **zero in 2025 or later**. The 2024 aggregate claim was published with a baseline panel whose knowledge cutoff predated most resolution dates. Any 2025-generation replication that uses current LLMs (Claude 4.7, GPT-5, DeepSeek Chat 2026) inherits a structural compromise: the model’s knowledge cutoff postdates every resolution. A 30-call probe on a uniform random sample from this dataset returned Brier 0.13 on 14 of 14 fired confident-NO cases with empirical YES rate = 0.07 – a calibration signature that is statistically indistinguishable from outcome recall on a forecasting task. We then ran a stricter source-currency panel on newly collected Manifold questions: 40 pre-cutoff and 40 post-cutoff contracts matched on source, topic, question-length bucket, and computed cutoff relation, with Claude, Codex 5.4-mini, and Gemini run tool-free on all 240 dispatch rows. The panel is 240/240 schema-valid. Post-cutoff rows have higher Brier than pre-cutoff rows by +0.191098 in aggregate; the paired-stratum delta is +0.2155 with permutation $p = 0.0004$ and 95% interval [0.1584, 0.2795]. A later partial repair joined 51 of the 80 contracts to pre-outcome probability metadata; 27 family/stratum/base-rate-band paired cells still show post-minus-pre Brier +0.255418, and an adversarial sensitivity check over the 29 unjoined rows leaves the effect positive at +0.127901. The same 51 pre-outcome Manifold probabilities provide a narrow market bar: mean Brier 0.099673

versus the joined LLM-panel mean 0.166963 overall, with the market ahead on post-cutoff rows and the LLM panel ahead on pre-cutoff rows. The matched evidence unit is the contract, not the market platform’s trader count and not the 153 LLM calls; an effective- N audit finds 17 YES / 34 NO outcomes, 32 of 51 market rows with Brier < 0.05 , and a post-cutoff market-vs-panel gap of 0.085272 versus 0.285093 with paired $p = 0.0034$. A simple market+LLM blend fails promotion: leave-one-out grid Brier is 0.097218 versus market-alone 0.099673, but the paired delta is only -0.002455 ($p = 0.794$, 95% interval $[-0.021, 0.0166]$), and the post-cutoff subset selects market-only. A follow-up void audit makes the scope explicit: the Stage-B LLM panel has 80 contracts / 240 calls, but only 51 contracts have an ingested market/human baseline, all Manifold, with 29 Stage-B contracts still unjoined. This is not a broad equal-information human/crowd baseline. The second-source replication remains open, although a public Polymarket CLOB-history probe now supplies candidate rows for the Polymarket acquisition slice; those rows have pre-outcome price history, full-pool max-one-row-per-event-family selection fills 33 of 33 target rows, final outcome prices agree with the decoded Yes/No outcomes for all 33, and all 33 are now DB-ingested under an explicit platform-resolver provenance tier. That tier uses the Polymarket resolved outcome plus market-page criteria text; it is not independent external-source verification. Subsequent Gemini and DeepSeek Polymarket smokes over 24 pre-cutoff and 24 post-cutoff matched rows were each 48/48 schema-valid. The aggregate direction reproduced for Gemini (post-minus-pre Brier $+0.246832$) and weakly for DeepSeek ($+0.077758$), but the six matched source/topic/length strata were null or opposite-sign (Gemini $+0.005731$, $p = 0.9696$; DeepSeek -0.061706 , $p = 0.8836$); this is useful stress evidence, not a source-general replication. A follow-up availability probe found that the natural Polymarket market-price control is not executable yet on the frozen slice: pre-cutoff rows have 24/24 DB freeze-price values, but post-cutoff rows have 0/24 locally and a live Gamma/CLOB probe joined 0/24 because the public endpoints reset connections from this environment. The remaining Metaculus target cells still require bot-benchmarking/data-download access or a licensed export: authenticated access to the current `/api/posts/` endpoint returns post/question payloads but not resolved Yes/No values or dated aggregate history for the sampled resolved binary rows, and `/api/data/download/` is restricted. Dataset-source rows such as FRED or yfinance can support a separate frozen source-currency replication design, but they are not a drop-in substitute for the existing Metaculus source/freeze-probability/length cells. A credential-aware FRED manifest audit on 2026-06-04 joined the frozen ForecastBench FRED question/resolution bundle to official FRED observations: 49 of 50 rows were scoreable, all 49 computed binary outcomes matched the bundled `resolved_to` values, and all scoreable rows were post-cutoff by resolution date. This proves post-cutoff official-data supply for a possible source-currency lane, not a pre/post replication, not manifest completeness, and not a human/market equal-information baseline. The original base-rate limitation is now a partial-repair limitation, not an absence of base-rate evidence. The deployable check before any replication is one filter: `resolve_date > max(panel_cutoff)`. That filter empties this dataset for the current LLM generation. The general point is corpus-validity drift: a benchmark that was honest at publication time may stop being honest once a new model generation has trained through the resolution dates. The verdict resolver discipline extends to corpus eligibility, not only to per-claim power.

The Stage-C market bar is now also materialized as typed external-baseline observations; all 51 rows have equal-information flag 0, so this remains a not-equal-information pre-outcome market comparison rather than a broad human/crowd baseline.

FRED update. A fixed one-year historical companion subsequently supplied 49 of 49 scoreable pre-cutoff FRED rows under the same official-observation rule. The resulting 49-series FRED pair packet ran Gemini and DeepSeek tool-free on 196 of 196 schema-valid calls, and the 98 contracts plus 196 calls are DB-ingested. The full paired post-minus-pre Brier delta is only $+0.016477$, with 90%

bootstrap interval $[-0.006445, +0.043330]$ and sign-flip $p = 0.30375$; DeepSeek is direction-positive but below the promotion bar, while Gemini is weaker. The model mean Brier 0.226225 beats a $p = 0.5$ baseline and the global empirical YES-rate baseline, but loses to the relation-specific empirical YES-rate baseline 0.213244, showing that outcome mix remains a central confound. This upgrades FRED from post-cutoff supply to an official-time-series stress lane, but not to a clean second-source Law 3 replication, not to manifest completeness, and not to a human/market equal-information baseline.

FRED confound control and demotion. A follow-up diagnostic packet selected 24 outcome-balanced FRED series, hid the cutoff relation from the prompt, and added a due-date-observed-value arm. Gemini and DeepSeek completed 192 of 192 schema-valid calls, now DB-ingested under `fred.blinded_value_control_v1`. On the current-FRED labels, the paired post-minus-pre Brier delta was $+0.024719$, with 90% bootstrap interval $[+0.007125, +0.043730]$ and sign-flip $p = 0.02315$. But a later no-call vintage timing audit showed those labels were not time-stable. The first `realtime_start/realtime_end` pass was rate-limited at 58 of 98 scoreable rows; a follow-up bulk repair used `FRED series/observations` with `output_type=2` and `vintage_dates`, batching once per series, and reached 98 of 98 scoreable rows with 49 of 49 series API-ok. Under label-time values, 49 of 98 due values changed as-of due, 44 of 98 resolution values changed as-of resolution, and 15 of 98 two-point binary labels changed. Rescoring the existing calls on the complete repaired label set gives pair-panel post-minus-pre Brier $+0.018721$ under vintage labels, with mean Brier worsening from 0.226225 current-label to 0.233266. The blinded-control apparent current-label penalty collapses from $+0.024719$ to -0.002989 . A reusable dataset-source label-time gate then audits the calibration DB and refreshes a row-level eligibility table: among 108 resolved dataset-source rows, 83 current-label rows are supported by available source-specific label-time receipts, while 25 are ineligible (15 FRED label changes plus 10 yfinance/yfinance_etf rows without label-time receipts). Therefore the current-label FRED positives are apparatus evidence, not positive Law 3 evidence. The next credible dataset-source continuation is an ALFRED/bulk-export confirmation, yfinance as-of/corporate-action receipts, or a third official-data source, not more same-shape prompt calls.

18.4 Lane B external proof audit: the helper-vs-top-level distinction and an explicit retraction

The verdict-resolver discipline extends to independent proof audits the program runs on other groups' published Lean artifacts. We audited eight bare-Mathlib4-importing top-level theorems published by an AlphaProof-line collaborator ("APN" set, Conjecture2 and proofs P1–P8) through a strict L1+L2+L3 stack (compile + axiom allowlist `{propext, Classical.choice, Quot.sound}` + anti-pattern gates against gold-name-verbatim leakage, single-lemma-exact paraphrase, simp/fun_prop indirect leakage, and scalar-wrapper currency mismatch). Four rounds of the audit reported successive "laundering caught" verdicts before we found the failure mode.

Retraction. The first three audit rounds collapsed *helper-lemma* anti-pattern hits into a *top-level* verdict, which is the wrong granularity: the top-level theorem statement may be clean even when an auxiliary lemma supporting it has a helper-level anti-pattern hit (e.g., naming a Mathlib lemma verbatim during a private intermediate step). We do NOT claim DeepMind published anything fake. The earlier "laundering caught" framing was wrong and is explicitly retracted in this draft. The corrected status rule distinguishes `compile_pass_l3_advisory_review` (clean) from `compile_pass_l3_advisory_review_helper_blockers_only` (top-level clean, helper-level flagged), and ONLY the former-or-this-latter pair counts as audit-clean. Confirmed top-level blockers remain disqualifying.

With the corrected rule plus a forced v4.27 pinned-toolchain sidecar for non-drift `compile_failed` cases (which disambiguates “unrecognized toolchain drift” from “real defect”) and a process-group kill discipline on lake subprocesses (the v33 audit was leaking orphan `lake` processes that ran 47–63 minutes past their timeout budget, blowing the run), the corrected verdict on the eight top-level theorems is **6 of 8 audit-clean**: Conjecture2 passes-at-native-toolchain via v4.27 sidecar; P1 advisory-review at v4.30; P2 advisory-review-helper-blockers-only at v4.30; P3 passes-at-native-toolchain via v4.27 sidecar; P7 passes-at-native-toolchain via v4.27 sidecar; P8 advisory-review-helper-blockers-only at v4.30. P4 and P5 are infrastructure-blocked at the time of submission: P4 has a `sidecar_invocation_failed` verdict from a candidate-path parsing bug in our harness rather than a science verdict; P5 is the same harness bug. Receipts at `analytics/public/queries/lane_b_apn_audit_receipts.json`. The methodological point is that an audit pipeline must be wrong four times before it gets the helper-vs-top-level distinction right; we report the four rounds and the retraction transparently because the same conflation pattern is a plausible failure mode in any third-party Lean audit.

18.5 The deployable consequence

The re-audit guardrail: *a per-family or per-(family, sub-source) routing rule should be promoted only when its supporting CI has been computed on $N \geq n_{\text{required}}$ at the target effect size, and only with sub-source stratification reported.* The pilot queue’s three remaining cheap retests ($\sim 0, 42$, and 420 calls respectively) close the same gap on the program’s other policy hypotheses; their results are published alongside the existing rules as the retests complete.

19 What this paper does NOT establish

- It does NOT show that cognitive-debiasing interventions never transfer to LLMs. Four of five tested are `inconclusive_underpowered` at our low-overlap-corpus N , not falsified.
- It does NOT establish that GPT-5.4-mini’s low-overlap-corpus frequency-framing improvement would survive cross-corpus replication at proper power. $N \geq 91$ public-domain is required.
- It does NOT generalize beyond five model families and two corpus classes. The low-overlap corpus is research-internal and not publicly available in raw form.
- It does NOT decompose LLM *reasoning* mechanistically. The channel-orthogonality claim (§8) is at the *expression* level; a reasoning-decomposition claim would need white-box mechanistic interpretability.
- The four-axis confounding between corpora means the cross-corpus result is evidence that at least one axis matters, not yet a clean attribution. The 4-cell de-confounded corpus is the most important methodological follow-up.
- It does NOT establish that RLHF/alignment *causes* the per-family bias differences (Axis 3). Every Axis-3 result, in-distribution and out-of-distribution, is a *cross-family* contrast, which confounds alignment with pretraining-corpus differences; families differ in both. The apparent “more alignment, less inherited bias” ordering also runs opposite to the established result that instruction-tuning amplifies several cognitive biases [4]. The honest Axis-3 claim is therefore the weaker, observational one—*family identity predicts bias magnitude*—not the causal *alignment damps bias*. Isolating the alignment stage requires within-family checkpoints (pre- vs post-RLHF) or matched-pretraining families, which we do not have.

- Bid-ask spread cross-corpus is currently underpowered; the direction-shift is suggestive but not conclusive at $N=42$.
- The worry-Brier sign-flip mechanism is provisional (§10 multi-probe synthesis); activation-level probing and a topic-matched corpus would settle it.
- Per-family signal strength is conditional on prompt-invariance (§16). Gemini and DeepSeek findings should be read with the prompt-stability discount documented there until the within-family paraphrase pilot is run.

20 Reproducibility

All pilot calls persist as JSONL files and are ingested into a SQLite database at `analytics/public/calibration/for`. Pre-registrations and verdict resolutions are in the same database. The reusable general-purpose statistics module is at `src/ztare/experiment_stats.py` (power calculator, bootstrap CI, paired permutation, Fisher- z Spearman, TOST, BH-FDR, power-aware verdict, BIC Bayes factor, reproducibility hash). Forecasting-specific wrappers are in the per-project `calibration_stats.py`. Every reported finding reproduces via CLI: `calibration_stats.py finding frequency-framing, brier-ci --primitive baseline --corpus low-overlap, spearman --pilot bid-ask --x spread`, etc.

Low-overlap-corpus reproducibility. The low-overlap corpus is research-internal and not publicly releasable in raw form. Remediation paths: (a) sanitized release with research-internal identifiers replaced by neutral tags (e.g., “Operation X tactic Y on row class Z ”), preserving the four-axis structural profile; release pending clearance. (b) Parallel public alien-domain corpus drawn from public-but-niche academic sub-fields (specific Lean mathlib lemma classes; obscure subfields of partial differential equations; verifiable but low-traffic historical-event resolutions) matching the low-overlap corpus’s four-axis profile, intended for direct third-party replication of the frequency-framing and bid-ask-spread results.

References

- [1] Hal R. Arkes and Catherine Blumer. The psychology of sunk cost. *Organizational Behavior and Human Decision Processes*, 35(1):124–140, 1985.
- [2] Shane Frederick, George Loewenstein, and Ted O’Donoghue. Time discounting and time preference: A critical review. *Journal of Economic Literature*, 40(2):351–401, 2002.
- [3] Danny Halawi, Fred Zhang, Chen Yueh-Han, and Jacob Steinhardt. Approaching human-level forecasting with language models. *arXiv preprint arXiv:2402.18563*, 2024.
- [4] Itay Itzhak, Gabriel Stanovsky, Nir Rosenfeld, and Yonatan Belinkov. Instructed to bias: Instruction-tuned language models exhibit emergent cognitive bias. *Transactions of the Association for Computational Linguistics*, 12:771–785, 2024. arXiv:2308.00225.
- [5] Daniel Kahneman and Amos Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.
- [6] Daniel J. Korchinski, Alessandro Favero, and Matthieu Wyart. Learn from your own latents and not from tokens: A sample-complexity theory, 2026.

- [7] Tamera Lanham, Anna Chen, Ansh Radhakrishnan, Benoit Steiner, Carson Denison, Danny Hernandez, Dustin Li, et al. Measuring faithfulness in chain-of-thought reasoning. *arXiv preprint arXiv:2307.13702*, 2023.
- [8] Sarah Lichtenstein, Baruch Fischhoff, and Lawrence D. Phillips. Calibration of probabilities: The state of the art. *Decision Making and Change in Human Affairs*, pages 275–324, 1977.
- [9] Philipp Schoenegger. Wisdom of the silicon crowd: Llm ensemble prediction capabilities rival human crowd accuracy. *arXiv preprint arXiv:2402.19379*, 2024.
- [10] Henri Tajfel and John C. Turner. An integrative theory of intergroup conflict. In William G. Austin and Stephen Worchel, editors, *The Social Psychology of Intergroup Relations*, pages 33–47. Brooks/Cole, 1979.
- [11] Philip E. Tetlock and Dan Gardner. *Superforecasting: The Art and Science of Prediction*. Crown, 2015.
- [12] Richard Thaler. Toward a positive theory of consumer choice. *Journal of Economic Behavior and Organization*, 1(1):39–60, 1980.
- [13] Amos Tversky and Daniel Kahneman. Belief in the law of small numbers. *Psychological Bulletin*, 76(2):105–110, 1971.
- [14] Amos Tversky and Daniel Kahneman. Judgment under uncertainty: Heuristics and biases. *Science*, 185(4157):1124–1131, 1974.
- [15] Amos Tversky and Daniel Kahneman. Advances in prospect theory: Cumulative representation of uncertainty. *Journal of Risk and Uncertainty*, 5(4):297–323, 1992.

A Composed routing recipe: closed form

Let $\bar{p}$ be the mean of the five families’ point estimates p_{success} on a given contract, h the days-to-resolution-from-cutoff, and $s \in \{\text{polymarket}, \text{manifold}, \text{yfinance}\}$ the source identifier. The composed routed forecast $\hat{p}$ is:

$$\begin{aligned}
 \hat{p}_{\text{conf-NO}} &= \begin{cases} 0.35 \bar{p} + 0.65 (0.10) & \text{if } \bar{p} < 0.20 \\ \bar{p} & \text{otherwise} \end{cases} \\
 \hat{p}_{\text{YES-bias}} &= \hat{p}_{\text{conf-NO}} + \begin{cases} +0.06 & \text{if } 0.30 \leq \hat{p}_{\text{conf-NO}} \leq 0.55 \\ 0 & \text{otherwise} \end{cases} \\
 \hat{p}_{\text{horizon}} &= 0.5 + (\hat{p}_{\text{YES-bias}} - 0.5) \cdot \frac{1}{1+0.01h} \\
 \hat{p} &= 0.5 + (\hat{p}_{\text{horizon}} - 0.5) \cdot w_s
 \end{aligned}$$

with $w_s = 0.70$ for polymarket, 0.85 for manifold, 1.00 for yfinance. The four coefficients (0.10, 0.06, 0.01, w_s) are not learned from held-out folds; they are chosen from the universal patterns reported in §14 (confident-NO over-confidence, middle-band YES under-shooting, horizon-conditional Brier slope, per-source Brier ordering). The per-channel routed alternative (routed.v2 in the main text) substitutes per-family channel weights for the universal w_s ; its weights are listed in the appendix companion file. The implementation reproduces from `projects/forecaster_skill_calibration_v1/workspace/c` on the database snapshot dated 2026-05-28.